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Start-up accelerator value chain: a systematic literature review

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Abstract
Accelerators are the crucial foundations that influence development of the entrepreneurial ecosystem and new venture creations. Although about 10 years have passed since the establishment of the first accelerator in the United States, there are still many dark spots about the accelerator's operations and services as well as their position and value in the entrepreneurial ecosystem. In this study, an attempt has been made to systematically review the articles published in the field of start-up accelerators to shed light on the dark spots. To this end, 51 articles and review articles published in the Q1 and Q2 journals of the Scopus citation database, were systematically and deeply studied. Parts of the main results of this research include the process of acceleration, the accelerator's services, the investing framework of accelerators, the success and risk factors of accelerators, and eventually antecedents as well as consequences of an accelerator. Finally, inspired by the Porter's value chain model, the accelerator value chain is presented as the final model of this study.

Keywords Startup accelerator · Entrepreneurship · Startup · Entrepreneurial ecosystem · New venture creation

JEL Classification M13 · L26

1 Introduction

Business Accelerator, Start-up Accelerator, and Seed Accelerator are the concepts which are relatively nascent (Busulwa et al. 2020). An accelerator is an organization or a business that helps start-ups to get through the difficulties of creating a new venture in the early stages and successfully launch it to market. Accelerators in general

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are the vital elements, which significantly raise the rate of new venture creations in the entrepreneurial ecosystem (Mohammadi and Shafiee 2021). Currently, there are many accelerators around the world, but their starting point goes back to the United States. Y-Combinator, the first world's accelerator, was founded by Paul Graham in Massachusetts in 2005 (Charoontham and Amornpetchkul 2021). This accelerator was a new generation of incubators that has had significant impacts on the innovation as well as entrepreneurial ecosystems by its new services (Cohen and Hochberg 2014; Drori and Wright 2018; Mohammadi and Heshmati 2021). After the success of Y-Combinator, David Cohen and Brad Feld, two start-up investors, founded TechStars in Colorado in 2007. These two accelerators solely helped just 2,000 start-ups to set up and raised more than $16 billion in capital (Cohen et al. 2019a, b; Cohen and Hochberg 2014). At present, the exact number of accelerators in the world is not available, but can be estimated at more than a thousand. Accelerators were not limited to a special field and were public in the early years of initiation, but in the recent years, acceleration programs have shifted to a focus on specialized fields in specific industries (Cohen and Hochberg 2014).

Some researchers consider accelerators as the incubators or new generation ones. (Gliedt et al. 2018), while others acknowledge that the main feature of accelerators separating them from other intermediaries such as incubators, is to provide targeted services in a limited time by an intensive program (Crișan et al. 2019; Miller and Bound 2011; Pauwels et al. 2016). Cohen introduces accelerators as a cohort-based program offering a variety of services, such as mentoring and training courses leading to the demo-day public presentation event (Cohen and Hochberg 2014). Cohen's definition is one of the most popular definitions among others and is the main basis of other definitions, which are discussed below. In a comprehensive definition provided by Dori and Wright (2018), an accelerator has been introduced as an organization whose main goal is to stimulate and motivate entrepreneurship (Drori and Wright 2018). It offers a limited-period acceleration program that includes training programs, mentoring, and networking (Tripathi and Oivo 2020). The start-ups participating in this program, in addition to increasing their capabilities to create a new venture, will also have the opportunity to participate in the demo-day event and the chance to attract capital from investors (Drori and Wright 2018). According to the various definitions provided by the researchers, it can be stated that accelerators directly affect the entrepreneurial tendencies (Del Sarto et al. 2020; Gutmann et al. 2020; Mahmoud-Jouini et al. 2018; Stayton and Mangematin 2019), start-ups' creation and growth (ventures) (Brown et al. 2019; Malek et al. 2014; Pandey et al. 2017; Pauwels et al. 2016), job creation (Breznitz and Zhang 2019), companies' growth (Brown and Mawson 2016; Gabrielsson et al. 2018; Shankar and Shepherd 2019), development of innovation (Coste and Gatzke 2017; Kupp et al. 2017), and the level of entrepreneurial knowledge (Cuvero et al. 2019; Del Sarto et al. 2020). By affecting these factors, accelerators lead an economic development for countries (Brown and Mawson 2016; Radojevich-Kelley and Hoffman 2012). Given the importance of accelerators' impact on the sustainable economic development (Cuvero et al. 2019), many researchers have done research in this area. The trend of the accelerators had an upward trend in recent years. These studies considered accelerators from a variety of perspectives, each examined some specific subject related to accelerators. The

scattering and non-integration of research views on accelerators as well as the scarcity of a practical guide for accelerators themselves, also, a need of a comprehensive model of accelerators' activities for the use of governments are the considerable gaps in previous research (Crișan et al. 2019; Drori and Wright 2018; Qin et al. 2019). In addition to this dispersion, the complexity of explaining this element in countries' entrepreneurial ecosystem is the another challenge (Cohen and Hochberg 2014; Stam and Van de Ven 2021). In general, it can be stated that the study gap expressed in the previous research (Cohen et al. 2019a, b; Crișan et al. 2019; Drover et al. 2017; Qin et al. 2019; Roundy 2017), face us with two fundamental questions: What is a start-up accelerator and what activities and processes does it constitute? What is the position and value of an accelerator in the entrepreneurial ecosystem?

Although, there are some studies that addressed to the gaps simultaneously (Bagnoli et al. 2020; Crișan et al. 2021; Yang et al. 2018), this study articulated all information about the accelerator's process and services. Also, all components that effect on and affected by an accelerator program are presented in the form of Porter's value chain. Moreover, preferences of investing on start-ups, transferring start-up's equities, success and risk factors of accelerators, types of accelerators along with their definitions are presented and can be a practical guidance for introduction of an accelerator or redesigning an existing one. On the other hand, the comprehensive model presented in the form of Porter value chain, would be a roadmap for future research in the field of accelerators' studies.

2 Methodology

The research method used in this paper is the systematic review (Buchheim et al. 2019; Marzall et al. 2021; Mumani et al. 2021). This method, which has entered from fields of medical sciences, has been widely used in various fields and has attracted the interest of many researchers (Boiko 2021; Röhm 2018). The systematic review has also been well received in the Management field in the recent years, and has been used in a variety of areas, including leadership (Eva et al. 2019), innovation (Rhaiem and Amara 2019), human resources (Pariona-Cabrera et al. 2020), radical innovation (Stiller et al. 2021), innovation portfolio management (Eckert and Hüsig 2021), digital transformation (Nadkarni and Prügl 2021) and many others. The systematic review is a structured research method for extracting knowledge from unstructured data. This method is generally summarized in three main steps (Tranfield et al. 2003) have been done as following.

2.1 Review planning

At this stage, the research scope and review protocol are determined. This protocol includes information such as the source of publication, the year of publication, the classification of articles, the rank of articles, and many other similar items that are required to develop a search strategy (Petticrew and Roberts 2008). In this study, the subject of examination is accelerator. As the Scopus database currently is the

leading indexed database and includes high ranked articles which are accessible to almost all universities, it was chosen as the source database. To this end, all articles published in Q1 or Q2 journals of the Scopus citation database were reviewed. Only two categories of "article" and "review article" were selected for this step.

The keywords were used to adjust the query of the search, were "entrepreneurial accelerator", "start-up accelerator", "seed accelerator", and "business accelerator to consider a range of alternative vocabularies arrangement. The results of this query concluded 496 articles that their titles, abstracts, and contents were reviewed using PRISMA method in several consecutive stages (Moher et al. 2009) and eventually resulted in 42 final articles. Following the finalization of these articles, by reviewing backward and forward in the references of all articles, 9 more articles were added to these articles, which have finally become 51 selected articles. Details of these articles are given in the Fig. 1.

2.2 Review execution

This step involves collecting the basic and haphazard data from the articles, processing, categorizing, and synthesizing respectively which resulted in the structured data. For this purpose, Excel software was used, so each row and column was assigned to the specific information extracted from articles. The meaningful information that was extracted from the 51 selected articles, was grouped in: process of publishing articles, research methods, process of acceleration, accelerator's services, initial capital and share ownership percentage by accelerators, key factors in accelerator's success, accelerator's risks, and much more. All information was derived from a comparison of the similarities and differences between the 51 articles.

2.3 Reporting

In this step, the report on the two previous stages is fully described and the analysis of the results and findings is fully presented accordingly. Table 1 provides information on each of the 51 reviewed articles, including article's title, year of publication, journal title, journal classification, citation number of articles, research questions, research methodology, and the number of cases/samples per study. NULL values were related to articles which their case number were not evident. Codes are assigned to each article as preventing text's length.

3 Results and analysis

In this section, the examined articles are described in detail. In order to analyse the results, all selected articles have been thoroughly studied and entered in Excel software as semi-structured data. Then, by several stages of structuring the data, structured information related to each article are prepared and the results are obtained. These steps have been re-examined by another researcher to increase the accuracy

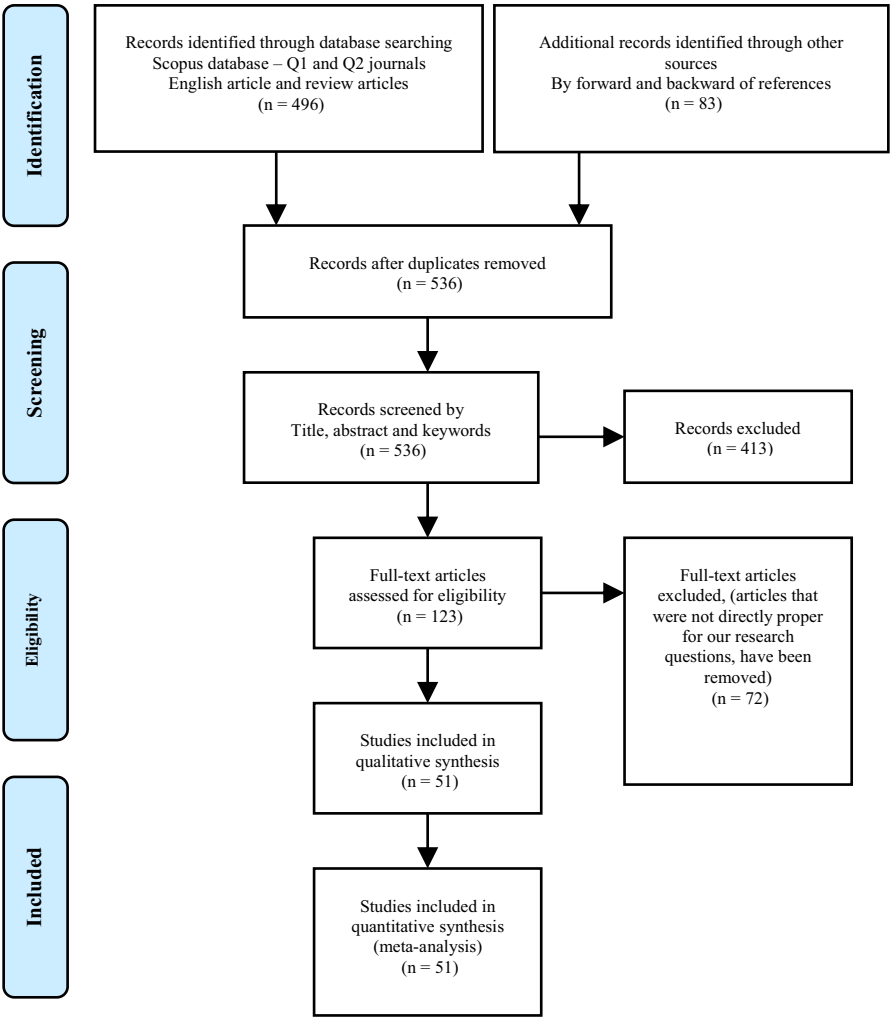


Fig. 1 Systematic review selection and review procedure adapted PRISMA 2009 (Moher et al. 2009)

of data entry. As described in the previous section, a total of 51 articles have been selected for systematic review from the Scopus citation database. The final Excel contained 51 rows of data (each row belongs to an article) and 38 columns (structured information of attributes), which eventually constituted 1938 cells.

3.1 Trend of studies

The articles selected in this study are all from 2014 onwards, which stands for the emerging trend of the research field. In addition, the upward trend of published articles in this area indicates an increase in researchers' attention to this crucial

Table 1 Detail of structured data of selected articles for systematic review (own research)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
1	(Del Sarto et al. 2020)	Technovation	Q1	0	1. Is there a relationship between participating accelerator programs and the firm survival? 2. How does participating accelerator's program interact with firm-survival variables? 3. Are variables related to firm survival the same as accelerators and incubators?	Quantitative	38 accelerated start-ups from five Italian accelerators and a control group of 38 non-accelerated Italian start-ups
2	(Breznitz and Zhang 2019)	Industrial and Corporate Change	Q1	3	1. How does managing an accelerator by who is a habitual entrepreneur affect firm's growth? 2. How does an accelerator program affect firm's growth? 3. How does screening process of an accelerator affect firm's growth?	Mixed	11 interviews with 9 accelerators in the university of Toronto
3	(Brown et al. 2019)	European Planning Studies	Q1	6	What role do accelerators play in mediating the process of transnational entrepreneurship?	Qualitative	8 Narrative interviews by entrepreneurs that selected for acceleration program
4	(Cohen et al. 2019a, b)	Research Policy	Q1	1	What are the heterogeneous design elements of accelerator programs?	Quantitative	146 accelerator programs, obtained from the Seed Accelerator Rankings Project

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
5	(Crișan et al. 2019)	The Journal of Technology Transfer	Q1	0	1. How different types of accelerators are going to be operated? 2. What outcomes do accelerators produce across different contexts?	Literature review	98 papers
6	(Cuvero et al. 2019)	IEEE Transactions on Engineering Management	Q1	0	What effects do knowledge spillovers and accelerator programs have on the product innovation of high-tech start-ups?	Qualitative	32 semi-structured interviews of CEOs and Cofounders of accelerated start-ups
7	(Mansoori et al. 2019)	Technovation	Q1	2	1. What effects do coaches who employed lean start-up methodology have on accelerators? 2. How does employing lean start-up methodology in an accelerator influence the relationships between entrepreneurs and coaches?	Qualitative	10 entrepreneurial teams and 6 coaches in an accelerator's program
8	(Moschner et al. 2019)	Business Horizons	Q1	3	What factors make a corporate accelerator?	Qualitative	20 interview 12 managers of 12 selected corporate accelerators

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
9	(Qin et al. 2019)	Industrial and Corporate Change	Q1	2	1. How do entrepreneurs in the accelerator ecosystem organize their activities to speed up the entrepreneurial process? 2. How does the time compression addressed by an accelerator ?	Qualitative	One cohort of start-ups in a leading accelerator in China
10	(Tripathi et al. 2019)	Information and Software Technology	Q2	1	How do start-up ecosystem elements affect MVP development in software start-ups?	Qualitative	13 start-ups and five supporting organizations (accelerators, incubator, co-working space, and investment firm)
11	(Yang et al. 2020)	Journal of Business Venturing	Q1	1	What is the influence of gender on selection decision of social accelerator?	Literature review	17 papers
12	(Bliemel et al. 2019)	Entrepreneurship and Regional Development	Q1	6	1. How do accelerators play an infrastructure role at the cluster level? 2. How do accelerators provide tangible and intangible dimensions of a start-up infrastructure to form a cycle of entrepreneurial activities? 3. How do accelerators develop start-up infrastructure?	Qualitative	28 interviews by 4 cases of accelerators

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
13	(Bustamante 2019)	Journal of Business Research	Q1	4	What is the best decision for outsourcing in accelerated start-ups?	Quantitative	261 accelerated start-ups
14	(Cohen et al. 2019a, b)	Administrative Science Quarterly	Q1	6	What is the best structure for accelerators in contributing new ventures?	Mixed	40 accelerated start-ups
15	(Prexl et al. 2019)	R and D Management	Q1	1	What are the drivers of heterogeneity among ecosystem builder accelerators?	Qualitative	20 leading experts from 16 German corporate accelerator
16	(Shankar and Shepherd 2019)	Journal of Business Venturing	Q1	6	How to run and structure a corporate accelerator?	Qualitative	4 cases of corporate accelerators
17	(Stayton and Mangematin 2019)	Journal of Technology Transfer	Q1	2	1. What are the underlying mechanisms allowing new ventures to speed up new product development and establish a viable organization simultaneously? 2. How do accelerators effectively support nascent firms to start up quickly?	Qualitative	8 U.S. accelerator programs and 37 accelerated start-ups
18	(Durda and Ključnikov 2019)	Economics and Sociology	Q2	0	How is the relationship between the type of nodes in the network and how they help?	Quantitative	100 start-up founders

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
19	(O'Dair and Owen 2019)	International Journal of Entrepreneurship and Innovation	Q2	1	Does blockchain technology offer a new added value to ventures, or merely reinvent existing corporate structures?	Qualitative	3 accelerated start-ups
20	(Al Taji and Bengo 2019)	Journal of Social Entrepreneurship	Q2	0	How do the distinctive managerial challenges of hybrid organizations appear in practice? and which skills can be taught to respond to those challenges?	Qualitative	3 social incubators and accelerators 6 start-ups
21	(Gliedt et al. 2018)	Journal of Cleaner Production	Q1	0	How do innovation intermediaries interact with other important actors and regime actors, to contribute a sustainability?	Literature review	20 papers
22	(Goswami et al. 2018)	Strategic Entrepreneurship Journal	Q1	23	1. How do accelerators affect venture success or failure? 2. How do accelerators affect regional entrepreneurial ecosystems?	Qualitative	20 in-depth semi-structured interviews by accelerated start-ups

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
23	(Mahmoud-Jouini et al. 2018)	Research Technology Management	Q1	4	What are key factors in building a corporate accelerator capability?	Qualitative	46 interviews by 22 Founders of accelerated start-ups, 10 Accelerator Program Managers from six accelerators, 12 Mentors and 10 Entrepreneurial Education Conference participants
24	(Yin and Luo 2018)	IEEE Transactions on Engineering Management	Q1	2	1. What critical criteria differentiate the selected start-ups from the rejected ones? 2. Do critical criteria vary across different decision stages of the start-up selection process? And if so, how?	Qualitative	35 semi-structured interviews by 8 accelerator managers and 4 accelerated start-ups
25	(Clayton et al. 2018)	Academy of Management Perspective	Q1	0	How do intermediary organizations facilitate science commercialization through entrepreneurship?	Literature review	NULL
26	(Gonzalez-Uribe and Leatherbee 2018)	Review of Financial Studies	Q1	12	Do business accelerators affect new venture performance?	Qualitative	20 women entrepreneurs and founders of accelerated start-ups
27	(Richter et al. 2018)	Creativity and Innovation Management	Q1	5	What are the essential and non-essential features of corporate accelerator programs?	Quantitative	3,258 start-ups (616 accelerated and 2,642 non-accelerated)

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
28	(Gabrielsson et al. 2018)	Journal of Cleaner Production	Q1	0	What is the effect of networked acceleration of innovation?	Mixed	19 CEO of accelerated start-ups
29	(Shane and Nicolaou 2018)	Journal of Institutional Economics	Q1	3	How technological advance, specifically the decreased cost of software developement, has driven a shift in the financing companies?	Qualitative	11 interviews with 6 managers of accelerator programs and 5 accelerated start-ups
30	(Connolly et al. 2018)	International Food and Agribusiness Management Review	Q2	2	How to implement and coordinate a corporate accelerator?	Qualitative	11 interviews with CEO of accelerators, 17 founders of accelerated start-ups and 3 other experts in 4 cases of corporate accelerators
31	(Seet et al. 2018)	Asia Pacific Business Review	Q2	5	1. How does human capital used in the education department of accelerators? 2. How social capital adds value to the process of entrepreneurial learning?	Qualitative	20 interviews by accelerated start-ups, non-accelerated start-ups, and mentors
32	(Yang et al. 2018)	Entrepreneurship Research Journal	Q2	3	Where each type of accelerator fits in the ecosystem and how they benefit their targeted start-ups?	Literature review	17 papers

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
33	(Drover et al. 2017)	Journal of Management	Q1	64	What are key considerations in moving forward a venture financing?	Literature review	NULL
34	(Kuschel et al. 2017)	Cross Cultural and Strategic Management	Q1	14	What are the challenges for Women Start-up Founders in Latin American?	Qualitative	14 accelerated start-ups
35	(Roundy 2017)	International Entrepreneurship and Management Journal	Q1	21	What is the role of support organizations in entrepreneurial ecosystem?	Qualitative	NULL
36	(Wright et al. 2017)	Journal of Technology Transfer	Q1	45	What factors characterize an entrepreneurial ecosystem and what role these factors play in the success of student start-ups?	Literature review	NULL
37	(Battistella et al. 2017)	European Journal of Innovation Management	Q2	15	How can start-ups benefit from participation in an accelerator program from an open innovation perspective?	Qualitative	Tracking a 12-week accelerator program, with 8 selected start-ups
38	(Jackson and Richter 2017)	International Journal of Innovation Management	Q2	8	What are the divergences in the interests of firms and start-ups that might lead to breakdowns?	Qualitative	27 interviews with accelerator managers and accelerated start-ups
39	(Miles et al. 2017)	Education and Training	Q2	7	How accelerators' activities facilitate the development of entrepreneurial capabilities?	Quantitative	77 accelerated start-ups

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
40	(Kupp et al. 2017)	Journal of Business Strategy	Q2	4	How to respond challenges caused by start-up for traditionally firms?	Qualitative	2 cases of corporate accelerators
41	(Coste and Gatzke 2017)	New Space	Q2	2	How is the effect of context on innovation?	Literature review	NULL
42	(Pandey et al. 2017)	Journal of Social Entrepreneurship	Q2	0	Does a social entrepreneur shape different benefits for a social accelerator?	Quantitative	48 social enterprise accelerator programs operated by 18 accelerators
43	(Wright 2017)	Journal of Business and Finance Librarianship	Q2	0	How do entrepreneurs obtain finance?	Literature review	NULL
44	(Brown and Mawson 2016)	Environment and Planning C: Government and Policy	Q1	0	What is the rationale of growth program of accelerators and how effective they are in facilitating rapid growth of firms?	Qualitative	50 accelerated start-ups
45	(Kohler 2016)	Business Horizons	Q1	56	How to set up or enhance corporate accelerators?	Qualitative	40 interviews with managers and accelerated start-ups in corporate accelerators
46	(Pauwels et al. 2016)	Technovation	Q1	111	How do accelerators operate as a new generation incubation model and how do they differ from existing incubation mechanisms?	Qualitative	13 accelerators across Europe

Table 1 (continued)

Code	Reference	Journal title	SJR Quartiles	# Of citation	Research question	Methodology	Number of cases/samples
47	(Hochberg 2016)	Innovation Policy and the Economy	Q2	32	1. What are accelerators? 2. What effect do accelerators have on the regional entrepreneurial environment?	Literature review	NULL
48	(Meskhi et al. 2016)	European Research Studies Journal	Q2	1	What are the principles and models of developing a regional accelerators based on technical universities aimed for innovative start-ups?	Qualitative	NULL
49	(Cohen and Muñoz 2015)	Organization and Environment	Q1	0	How can the theory of place-based entrepreneurship, change to theory of purpose-driven?	Qualitative	20 accelerated start-ups
50	(Kim and Wagman 2014)	Journal of Corporate Finance	Q1	14	What is the role of a start-up accelerator in choosing a portfolio size and disclosing information about participating ventures?	Mixed	NULL
51	(Malek et al. 2014)	Journal of Engineering and Technology Management—JET-M	Q1	24	How varied are core capabilities of an accelerator?	Qualitative	6 accelerators

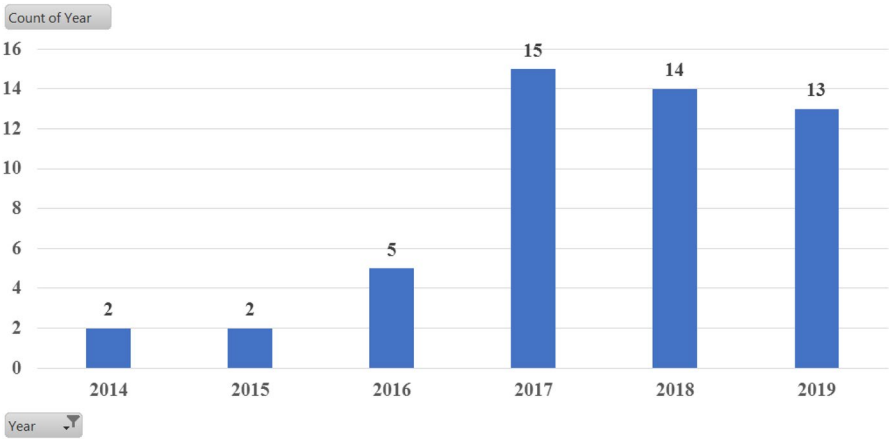


Fig. 2 Publication trend

foundation in the entrepreneurial ecosystem. The publishing trend of the reviewed articles is presented in the bar chart below (Fig. 2).

One of the most important pieces of information in the process of studying a field is the research method. For this purpose, the method of conducting research for selected articles has been categorized into four categories: qualitative, quantitative, mixed, and review, and the process of their changes in terms of time are reviewed. In general, the largest share of studies is related to qualitative studies, with 30 articles at the top of these four categories. After that, the review articles accounted for 10 cases, articles with qualitative research methods are 7 and articles with mixed research methods are 4 cases respectively. The Fig. 3, illustrates the trend of using each research method over the time. The qualitative research method had an increase among the rest methods.

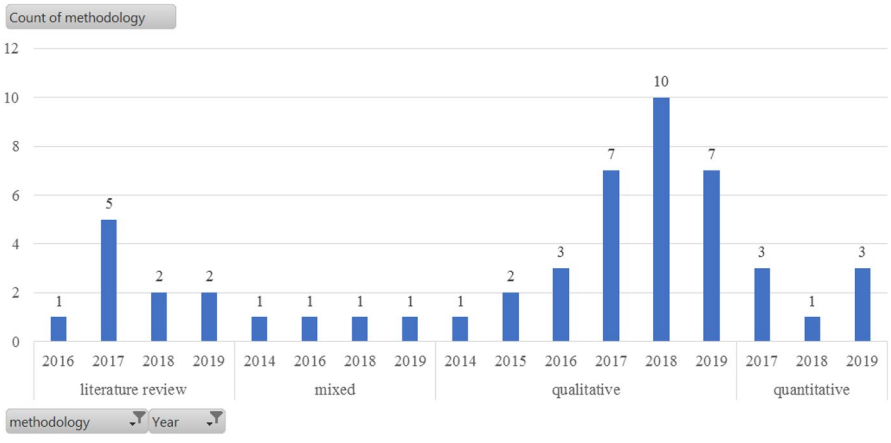


Fig. 3 Trend of research methodologies in selected articles (own research)

3.2 The journals

These articles were published in various journals with high H-index. The majority of these journals are in the areas of technology management and entrepreneurship management, and include well-known journals such as Technovation, Journal of Business Venturing, Journal of Technology Transfer, R&D management, Entrepreneurship Research Journal, and more. Among them, the journals which included two or more articles selected for this study are presented in the Table 2.

3.3 Types of accelerators

The description of an accelerator has many details leading to different types of definition for accelerators. Of course, the services and features associated with them in all types of accelerators are very close to each other, so that it will be somewhat difficult to understand the difference between them. In this section, by examining all types of accelerators mentioned in each reviewed article, an attempt has been made to determine the narrow border between these different types. In order to make an easier comparing of accelerators, Table 3 provides the title, definition, and reference article for each type of accelerator. Separating types of accelerators does not mean that they are not related to each other, but in some cases, they are subsets of some types. For example, for-profit and non-profit accelerators are a large category for different types of accelerators that accommodate all types of accelerators. Also, virtual accelerators are the type of accelerator that is a general category. This category may be a seed accelerator, corporate accelerator, social accelerator, or any other type of accelerator alone. Table 3 introduces all types of accelerators separately.

4 Discussion

4.1 Acceleration program

Start-up Accelerators require a regular and accurate acceleration program to transform the initial ideas of start-up teams to an available product in the market. These

Table 2 Top journal of accelerator scope

Source	H-index	Count
Technovation	111	3
Journal of Business Venturing	154	2
Journal of Cleaner Production	150	2
Industrial and Corporate Change	95	2
IEEE Transactions on Engineering Management	82	2
Business Horizons	67	2
Journal of Technology Transfer	66	2
Journal of Social Entrepreneurship	20	2

Table 3 Accelerator types (own research)

Accelerator types	Definition	Reference codes
Pre-accelerators	"pre-accelerators may involve hands-on teaching, so-called entrepreneurship garages as well as venture labs and co-working spaces"	36
Start-up Accelerator (SUA) or seed accelerator or business accelerators	These accelerators, which are the most well-known type of accelerator, are the general or common type of accelerator that provide services such as seed money, workspace, mentorship, training, networking, and more in 3 to 6 months. Their acceleration program ends with teams participating in DEMO-DAY and the chance to raise capital for angels and venture capitalists. These accelerators take a percentage of the start-up's stock in exchange for these services	4, 11, 17, 24, 27, 36
White-Label Accelerator (WLA)	Are designed to create an entrepreneurial ecosystem for one or more specific corporations. As a neutral facilitator, these accelerators accelerate the start-ups of these corporations	15
Idea-Lab Accelerator (ILA)	These accelerators have a laboratory space to support new ideas and technologies and future trends in specific areas of technology and focus on the edge of technology	15
Investor-led accelerators	These accelerators identify investment opportunities for private investors, including business angels, investment funds, venture capital firms, and more. These accelerators also focus on adult start-ups	26, 46
Intrapreneurship Accelerator (IPA)	This type of accelerator is formed to identify entrepreneurial opportunities by innovators and entrepreneurs within the organization	15
Growth accelerator	Are designed to grow start-ups quickly. In addition to seed money, they provide coaching services, rapid growth strategies	3
Corporate accelerator (5 types of Corporate Hackathons, Business Incubators, Corporate Incubation, Corporate Venturing and Mergers and Acquisitions)	These accelerators depend on a specific organization and meet the goals of the parent organization. These accelerators may operate within the parent company or as an independent unit. The main purpose of this group of accelerators is to identify relatively mature start-ups to meet the technical and specialized needs of the parent organization and provide their acceleration services. These accelerators can also be introduced as an open innovation platform for organizations	1, 8, 11, 23, 27, 30, 38, 40, 41, 45

Table 3 (continued)

Accelerator types	Definition	Reference codes
Ecosystem accelerator	These accelerators are designed to help start-ups grow and help identify target customers accurately. They also try to help the start-up ecosystem and entrepreneurship in their area by providing mentoring and coaching services without providing seed money and receiving stocks	26, 46
Social accelerators (a non-profit accelerator subset)	These accelerators aim to provide services such as seed money, training, consulting, networking, mentoring, access to investors, trying to help start-ups in social fields such as education, health, poverty, social inequality, unemployment, immigration, and other social challenges. These accelerators try to develop social entrepreneurship at the national and international levels in exchange for much less common stock	11, 46
For-profit accelerator (private accelerators)	This category of accelerators includes all types of accelerators that are formed for profit and stock acquisition purposes and profit from the success of start-ups and seek to increase their rate of return on investment	1, 11, 35, 36, 37
Non-profit accelerator (public accelerators)	This category of accelerators includes a variety of accelerators that are funded by public or private organizations and operate in line with social goals and social welfare development without expecting to receive stocks or with very small stocks from start-ups	11, 25, 35, 36
University- based	This category of accelerators includes accelerators that are supported by scientific and academic institutions and support start-ups for the purpose of economic development	11
Virtual accelerators	This category of accelerators are the ones which do not have a specific physical location or do not provide teams a shared workspace based on their period. All services of this type are virtual	28

programs are customized in each accelerator depending on the needs of start-ups and their ability of running plans. However, similarities of accelerators' programs are enormous and a general program can be put in place. In this section, all accelerators' programs presented in 51 selected articles were reviewed and a comprehensive program from the beginning to the end of accelerators' activities was presented. This program can be used as a guide for accelerators' executives, and each accelerator can design its program depending on needs and capabilities. The Table 4 presents 19 stages of an accelerator's program, that are divided into 5 main stages: planning, admission, pre-acceleration, acceleration, and post-acceleration. It starts with the accelerator's first step in defining the strategy of program then specifying the priority of scopes that are going to be called for start-ups' participation. Furthermore, steps proceed with providing teams to present to investors at DemoDay and finally post-launch programs to support accelerated start-up teams.

These programs take place at different time intervals, and each accelerator has a specific time frame for completing the program, depending on its strategy. Determination of the timeline varies from 1 month to 1 year depending on various factors such as the field of activity of start-up teams, the industry, accelerator's strategy,

Table 4 Acceleration program steps (own research)

Category	Steps	Reference codes
Planning	Determining the program's strategy	11, 15
	Determining the main scope of the accelerator's program to call teams	27
Admission	Calling for program's participation	27
	Initial registration of teams on website (initial proposal)	48
Pre-acceleration	Initial screening of teams	15, 27, 48
	pre-acceleration or boot camp	39
	Presenting ideas	7, 27
	Evaluating and judging teams and ideas to participate program	7, 11, 15, 27, 39
Acceleration	Seed money	9, 11, 39
	Holding different types of training courses for teams	5, 9, 27, 39, 46
	Providing specialized mentorship for teams	9, 27, 39, 46
	Providing expert advice for teams outside of the accelerator	9, 46
	Holding specialized events for teams	9
	Helping teams to develop a business model	39, 47, 48
	Helping teams prepare the minimum viable product for a start-up	39, 48
	Helping teams to validate the business model and the minimum viable product	5, 39
	DemoDay (networking and communicating of start-ups in the DemoDay event and possibility of fundraising by meeting opportunities)	7, 15, 27, 39, 46, 48
Post-acceleration	Rewarding teams and investing by investors on the start-up	9, 11, 15, 27, 46
	Holding support programs after program ended	46

and many other factors. To calculate the average time that required for an accelerator's program, this study reviewed articles from a time perspective and the Fig. 4 is based on time duration of programmes. The most frequented programme's durations belonged to 3-month and 6-month.

4.2 Accelerator services

Accelerators offer a wide variety of services to teams in the form of service packages presenting in the pre-acceleration, acceleration, or post-acceleration stages during program. These services range from providing initial capital for start-up, consulting, mentoring, and providing start-up visa for settlement of teams in foreign country accelerating start-up. Another finding from the review is the introduction of all possible services for accelerators. In the Table 5, all identified services are introduced in the form of 36 different services, which are divided into 6 main categories. These 6 categories include advisory services, business services, educational services, funding services, networking services, and office services.

4.3 The structure of initial investment

One of the services that is provided by accelerators is preparing the initial capital for running a start-up in the form of seed money. This amount is given to them in exchange for a percentage of the start-up's shares being acquired by the accelerator. The seed money is allocated in exchange of shares that will be received by the accelerator and varies depending on types of services, the scope of the accelerator activity, the duration of the acceleration program, and many other factors. The share percentage in the world, on average, is between 5 and 15 percent per 10 to 50 thousand

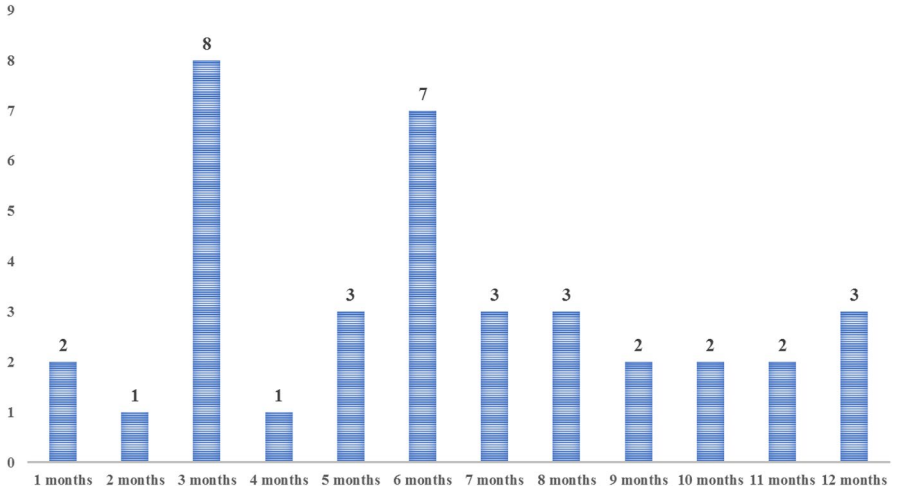


Fig. 4 Time durations of acceleration programs (own reserach)

Table 5 Accelerator services (own research)

Group	Service	Reference codes
Advisory services	Coaching	2, 5, 7, 23, 28, 39
	Consulting services (business, investment, legal etc.)	14, 17, 18, 19, 20, 37, 40, 47, 48
	Legal services	5, 13, 14, 37, 48
	Mentorship	1, 3, 4, 5, 9, 10, 11, 14, 16, 17, 19, 20, 22, 23, 24, 25, 26, 27, 29, 31, 33, 34, 35, 36, 37, 38, 39, 42, 43, 46, 47
Business services	Access to media and PR	12
	Accounting support	14
	Advertising on accelerator medias	5, 26, 48
	Bank account	5, 13
	Brand recognition	5
	Business support	3, 14, 16, 39, 46
	HR support (personnel selection, evaluation, personnel audit, preparation of development programs, etc.)	5, 22, 37, 48
	Idea generation and creativity management support	5, 8, 11, 30, 32, 36
	Increase credibility of start-ups	16, 42, 45
	Prototype and MVP building support	5, 8, 10, 11, 25, 32
Educational services	Bootcamps	5, 40
	Events, conferences, and workshops	5, 10, 20, 25, 26, 35, 38, 39
	Training and educational programs (entrepreneurial, financial, technical, and legal educations)	1, 4, 7, 9, 10, 11, 12, 13, 14, 16, 17, 20, 26, 27, 35, 36, 37, 39, 42, 46, 47
Funding services	A public pitch event or demo day to help attract capital	1, 5, 7, 14, 30, 32, 35, 37, 39, 47
	Access to round A funders	1, 3, 9, 12, 22, 25, 28, 29, 36, 42, 45, 47
	Pre-seed investment	37, 46
	Seed funding	1, 2, 3, 4, 5, 11, 12, 13, 16, 17, 19, 23, 24, 25, 26, 28, 29, 30, 33, 34, 39, 42, 46, 47
	Technical support	5, 12, 14, 25

Table 5 (continued)

Group	Service	Reference codes
Networking services	Access to a wide network of experts, investors, mentors, advisors, business, entrepreneurs, etc	1, 2, 3, 4, 5, 7, 9, 11, 12, 13, 14, 17, 18, 19, 22, 23, 24, 26, 27, 28, 30, 31, 32, 35, 37, 39, 42, 43, 46, 47
	Access to industry experts	14
	Access to market	14, 16, 36, 45
	Access to potential customers	1, 11, 14, 22, 24, 25
Office services	A full of passion competitive start-up ecosystem place	1, 5, 22, 37, 40, 46, 47
	Access to meeting room	17
	Access to physical resources (such as printer, scanner, boards…)	2, 5, 8, 14, 16, 18, 25, 27, 28, 45
	Access to R&D facilities and equipment	5, 7, 28
	Catering	5
	Free access to online platforms	5
	Getting a phone number	5
	Housing	5, 43
	Internet access	5, 13, 14
	Office space or co-working space	2, 4, 5, 7, 11, 12, 13, 14, 17, 20, 22, 26, 28, 33, 37, 39, 43
	Temporary one-year visa	13

dollars. In this section, by examining articles, the frequency of the share percentage obtained by accelerators as well as the amount of initial capital provided by them is identified. The Fig. 5 depicts the information.

As shown above, the highest repetition in the percentage of shares assigned to the accelerators is between 5 and 10%. One of the main factors influencing the determination of share percentage, is the amount of initial investments that are made by the accelerators. The following Table 6 shows the initial investment of accelerators, based on the reports presented in the reviewed articles.

4.4 Risks of accelerator

Risk recognition is one of the most important precautionary measures to prevent businesses' failures. An accelerator as an individual business model, subjects to the risks as well. Since an accelerator invests its limited financial capital and resources on a limited number of start-ups, it might be encountered many risks. In this section, it has been tried to outline the risks that are faced by accelerators. These risks are extracted from the context of the papers examined in this study and will be a feed for future researchers to be able to provide a framework for the risk management of accelerators through the active approach of risk management. Table 7 lists these risks.

4.5 Success factors of accelerators

This section identifies key success factors of accelerators. By paying attention and improving them, the success path of accelerators will be smoother. These success

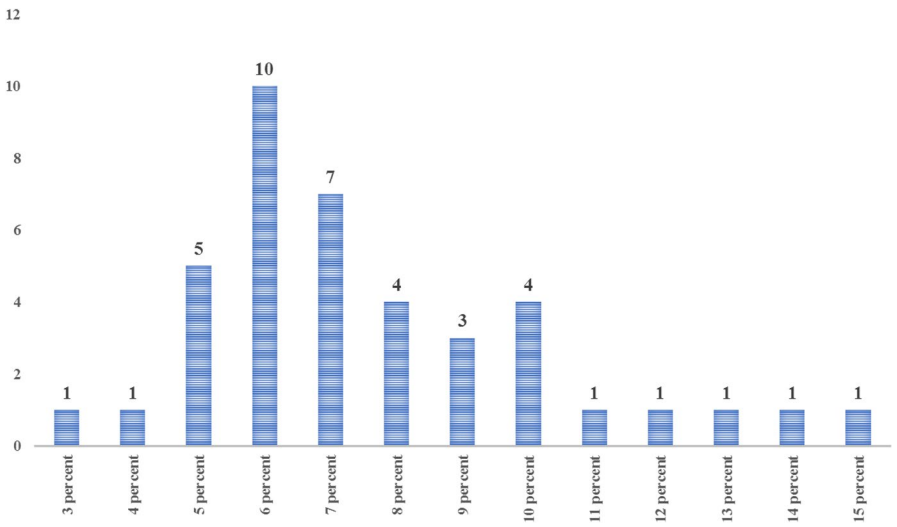


Fig. 5 Percentage of shares assigned to accelerator (own research)

Table 6 Seed money fund in accelerators

Reference codes	Seed money ($US)
34	$150 K–$350 K
40	Up to $300 K
45	$50 K–$150 K
4, 14	$50 K
9	$50 K
13	$40 K
32	$26 K on average, with a range from $0 to $150 k
47	$26 K on average, with a range from $0 to $150 k
33	$25 K–150 K
8	$25 K–$50 K
23	$25 K on average
24	$23 K
14	$15 K–$20 K in two- to three-person start-up teams
50	$11 K (plus $3 K per founder) in each participating venture
1	$10 K–$50 K
46	$10 K–$50 K
42	$5 K–$8 K per founder and between $18 K–$25 K per company

factors are presented in the 14 main factors in the Table 8. In the future research, researchers can use these factors to design and present assessment structures of accelerators' success.

4.6 Antecedent and consequence of accelerators

One of the most important questions in this research is to understand the factors that affect an accelerator as a concept. These factors may affect the accelerator in different ways and at different levels, so it will be very important to identify and manage them. Researchers have used a variety of qualitative and quantitative approaches to identify the factors that affect the accelerator. In this section, a total of 18 factors are recognised as an accelerator's antecedents which are shown in the Table 9.

On the other hand, an accelerator as a concept also affects different factors at different levels. 27 factors as accelerator consequences are identified based on reviewed article and presented in the Table 10.

5 Overview of accelerators

The purpose of this section is to provide a visual and conceptual model of all studies conducted in the field of accelerators, as well as a simpler understanding of how antecedents, consequences, and the process of an accelerator's programme are related together. Given that activities of an accelerator include a chain of core

Table 7 Risks of accelerators (own research)

Category	Risk of accelerators	Reference codes
Ideology differences	Contradiction of start-ups and accelerators in culture	11, 38, 45
	Contradiction of start-ups and accelerators in aims and objectives	11, 38
	Contradiction of start-ups and accelerators in personalities	11, 38
	Contradiction of start-ups and accelerators in beliefs	11, 38, 45
Operational skills and expertise	Finding expert mentors (especially in high-tech industry and developing countries)	12, 22
	Implementing program	5
	Managing the interaction with start-ups	23
	Managing accelerator	4, 14
	Selecting and filtering top-quality start-ups from a large number of applicants	24
Monetary and investing risks	Lack of capitals for investment	12
	Risks of Investment	8, 20, 23, 30, 45
	Lack of venture capital for post-acceleration funding	12
	Failure risk of the start-up	45

Table 8 Critical success factors of accelerators (own research)

Category	Success factors of accelerators	Reference codes
Strategic planning and managerial skills	Resource management	9, 12
	Innovation management and thinking skills	20, 30
	Long-term objectives and performance indicators	40
	Qualification of accelerator's program	2, 15, 46
	Lean methodologies	27
	A clear value proposition for the start-ups	15
Human capital characteristics	Management commitment	27, 40
	Dedicated business developers	23
	Complex combination of human capital in accelerator	47
Operational skills and expertise	Competitive start-up selection process	2, 24, 27
	Effective mentorship	12, 27, 30
	Quality of the selected start-ups	24
Networking	A large and committed external network	20, 30, 40, 47
	Ability to work with numerous start-ups	15
	Absorptive capacity of accelerator	15
	Access to investment	6
Infrastructure	Experience of accelerator	47
	Complementary resources	9, 30

Table 9 Antecedents of accelerator (own research)

Category	Antecedents of accelerator	Reference codes
Strategic planning	Public policy	44
	Strategic goals of accelerator	6, 15, 16
	Accelerator's model and program's package	2, 8, 15
	Funding structure	9, 15
Human capital characteristics	Background and Experience of accelerator's founders	4, 14
	Knowledge of investors	17
	Mentors	30, 31
Operational skills and expertise	Quality of accelerated start-ups in last cohorts	16
	Selecting process of start-up by accelerators	11, 15, 16
	Using lean start-up methodology in accelerator's program	7
Networking	Managers' networks	15
	Number of participated start-ups in accelerator's program	8
Ecosystem infrastructure	Innovation ecosystem components	3, 17
	Scope of Industry that start-ups are active in	6
Start-ups' characteristics	Commitment of start-ups	39

Table 10 Consequences of accelerator (own research)

Category	Consequences of accelerator	Reference codes
economic status	Sustainable economic development	6, 21
	Economic equilibrium	47
	Servitization	3, 31
	Making job opportunity	21
Entrepreneurial ecosystem	Development of entrepreneurial ecosystem	17, 22, 23, 25, 31, 32, 35, 37, 39, 47
	Building community between founders, investors, etc	24, 37
	Student entrepreneurship	2, 15
Innovation ecosystem	Open innovation	37
	Product and service innovation	6, 21, 30, 40, 41, 45,
Knowledge development	Knowledge acquisition	1
	Knowledge spill over	6
Start-up's supports	Access to funding opportunities for start-ups	19, 22, 24, 29, 30, 33, 47, 48
	Branding of accelerated start-ups	24
	Setting goals of accelerated start-ups	30
	Credibility of accelerated start-ups	24
	Start-up value	1, 50
	Enhancing knowledge of start-up founders	22
	MVP development for start-ups	10
Venture creation supports	New venture creation	6, 12, 13, 17, 23, 27, 28, 38, 39, 46, 47, 51
	New venture success	9, 22
	Venture capitals activities	33
	Venture growth	7, 16, 24, 37, 39, 42, 44, 47
	Venture performance	26, 49
	Venture strategies	30

and supportive activities, the Porter value chain model is used so that antecedents and consequences are presented as input and output of this chain (Porter 1985). (Details of each section are completely provided in the previous sections and here the macro model is shown. The Porter's value chain generally has two main parts: key activities and supportive activities. The process of an accelerator's programme is the core activity, which is divided into 5 general stages of planning, admission, pre-acceleration, acceleration and post-acceleration, and there are 36 services that identified as supportive activities, which are divided in 6 main categories including advisory services, business services, educational services, funding services, networking services, and office services. In this final model, the antecedents and consequences of an accelerator are also displayed in the form of out-of-chain factors, at both macro and micro levels, to provide an overview of the accelerator concept for researchers in this field as well as accelerator managers. This model is presented in the Fig. 6.

This model can be used as a guide for researchers, accelerator managers and policy makers in the entrepreneurial ecosystem. This model can show the systematic position of an accelerator in the entrepreneurial ecosystem of countries in addition to its general image of activities. Researchers can complete the structure of conceptual models of their research using this model, also, by different quantitative and qualitative methods can examine the impact of each factor on the accelerator and vice versa.

6 Conclusion and future research

The accelerator is a concrete element of the entrepreneurial ecosystem of every individual country, which is fundamental in creation of new ventures and growth of existing ones. This vital foundation which originates from Y-combinator in the world since 2010, still has many hidden angles that researchers and activists in the fields of entrepreneurship and innovation are required to discover. In this study, with the aim of revealing and discovering some of these hidden angles, a systematic and in-depth review of all research published in high quality journals has been conducted. For this purpose, a total of 51 articles and review articles published in the Q1 and Q2 journals of the Scopus citation database have been examined and discussed, and all their data has been gathered in Excel software as unstructured data. After several stages of structuring these data, eventually information about the research methods, case numbers, research questions, publishing trends, acceleration processes, services rendered, amount of initial funding, share percentages, different types of accelerators, programs' durations, key success factors, risk factors, and ultimately the antecedents and consequences of an accelerator concept are explored in this paper. Finally, the suggestions of the selected articles for the future research along with their results and limitations have been thoroughly examined so that based on the limitations and the results of the present study, to provide suggestions for qualified research in the future.

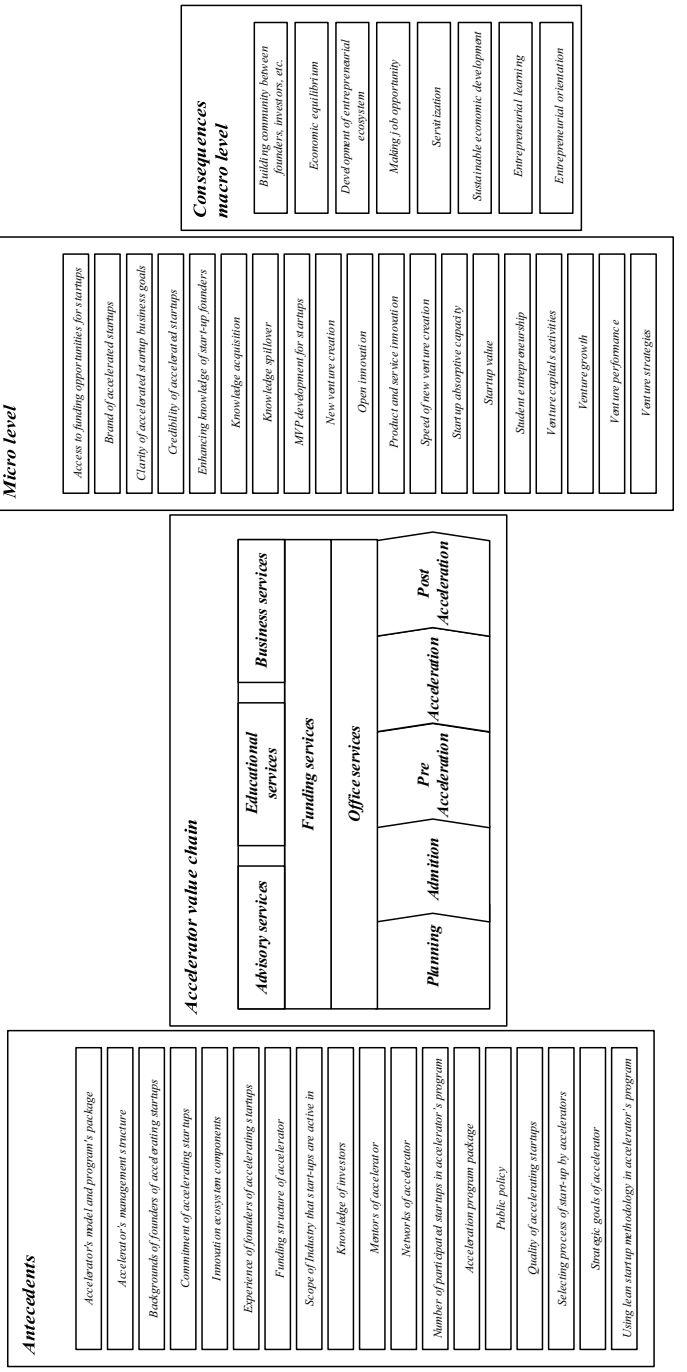


Fig. 6 Value chain of accelerators (own research)

6.1 Suggestions for scholars

Since the systematic perspective is one of the fundamental requirements of any scientific field, it is suggested to consider the antecedents and consequences identified in this research, to explore and theorize how factors affect each other along with their priorities as well as understanding the relationship of such factors with higher levels such as the entrepreneurial ecosystem.

The quantitative data-driven research based on multi-attribute decision making, data mining and machine learning is one of the emerging and evolving methods in the research.

According to the findings of this study, by multi-criteria decision-making methods, the factors which effect on and affected by accelerators as well as the risk and success factors of accelerators can be prioritized so that to provide risk and success assessment frameworks accordingly. Moreover, by machine-learning methods through gathering common features of the programmes, accelerators' programmes can be clustered. Also, by considering successful start-ups exited the programme as an anticipating factor, a structure for anticipating accelerators' success based on classification methods of machine-learning can be designed. These frameworks can be criteria for accelerators' assessment.

6.2 Suggestions for practitioners

Given the significance of selecting process of start-ups applying for program and the lack of research in providing a framework to guide accelerators in this process, a structure for assessment, scoring, and evaluating start-ups would be necessary and is recommended.

Funding There is no funding for this research.

Availability of data and materials All of data that used in this research is in a local excel file.

Code availability There is no code for this research.

Declarations

Conflicts of interest There is no conflict for this research.

Ethics approval There is no ethic issue for this research.

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